INCH-POUND

MIL-A-25546C (USAF)
AMENDMENT 2
26 AUGUST 1994
SUPERSEDING
AMENDMENT 1
6 April 1984

MILITARY SPECIFICATION
AMPLIFIER, AUDIO FREQUENCY AM-944/AIC-13

This amendment forms a part of Military Specification MIL-A-25546C (USAF) dated 18 June 1970 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

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2.1: Delete: Both "MIL-E-5272" and "MIL-C-8384" and its title.

* Delete: The title of MIL-STD-461 and substitute: "Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility".

* Delete: The title of MIL-STD-704 and substitute: "Aircraft Electric Power Characteristics".

* Delete: The title of MIL-STD-781 and substitute: "Reliability Testing for Engineering Development, Qualification, and Production".

Add: "MIL-STD-810, Environmental Test Methods" and "MIL-C-28748, Connector, Electrical, (Molded Body) and Accessories".

* Delete: The title of MIL-STD-810 and substitute: "Environmental Test Methods and Engineering Guidelines".

* Delete: The title of MIL-STD-28748 and substitute: "Connector, Electrical, Rectangular, Rack and Panel, Solder Type Pin Contacts, Size 16".

* Add: The headings "HANDBOOK" and "MILITARY" and list "MIL-HDBK-781, Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production"

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3.2.2: Delete "MIL-C-8384" and add "MIL-C-28748".

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* Add: "3.5.7.1 Talk-Listen Test. A talk-listen test shall be employed as specified in paragraph 4.5.6."

* Add: "3.5.9 Environmental Tests: Environmental testing shall be conducted as discussed in paragraph 4.5.7.

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* Table IV: Insert between Inspection or Test entries "Electrical Tests" and "Input-Output Characteristics Test":

Inspection or Test	Paragraph or Requirement	Preproduction	Individual	Sampling
"Impedance Tests	4.5.2.1	3		All"

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* Table IV: Delete the following Environmental Tests:
  Salt Spray
  Temperature Shock
  Sand and Dust
  Fungus
  and add under Environmental Tests (to be performed on 2 Preproduction amplifiers only):
  Salt Fog
  Explosive Atmosphere

* Table IV: Add under the following headings:

Inspection or Test	Paragraph o Requirement	Preproductio	Individual	Sampling
"Environmental Stress Screeni	4.5.9	all	all	all"

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* Add: "4.5.2.1 Impedance Tests. The amplifiers shall be tested for input impedances and output impedances as detailed in paragraphs 3.5.1 and 3.5.2, respectively."

* 4.5.3 Add at the end of the paragraph: "Linearity shall be tested for all amplifiers as specified in paragraph 3.5.3.1 and Table I. Clipping tests shall be performed on preproduction and sampling amplifiers per paragraph 3.5.3.2."

* 4.5.4 Add at the end of the paragraph: "The response is characterized in Table II"

* Add: "4.5.5.1 Power Drain. Power drain tests shall be performed as specified in paragraph 3.5.6."

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* Add: "4.5.5.2 Phasing. Phasing tests shall demonstrate the relationship described in paragraph 3.5.7."

* Add: "4.5.6.1 Operation. The amplifier shall perform as specified in paragraph 3.5.8."

* Add: "4.5.6.2 Electromagnetic Interference Test. The amplifier shall meet the characteristics for Electromagnetic interference control as specified in paragraph 3.2.4."

* Add: "4.5.6.3 Output Variation. The output of the amplifier shall remain within the bounds specified in paragraph 3.5.8."

4.5.7: Delete in its entirety and add: "Environmental Tests. The applicable environmental tests shall be performed in accordance with MIL-STD-810. Tests not listed do not waive the requirements in designing the component for airborne use or storage under extreme conditions. The following tests shall be conducted on the components of an intercommunication set:

a. Temperature-Altitude, Method 504, Procedure I
b. Humidity, Method 507, Procedure I
c. Salt Fog, Method 509, Procedure I
d. Explosive Atmosphere, Method 511, Procedure I
e. Vibration, Method 514, Procedure I, Curve M, Parts 1, 2, and 3
f. Shock, Method 516, Procedures I and II

* 4.5.8.1: Delete in its entirety and add:

"Reliability Test. The preproduction samples of the equipment shall be tested by the contractor for reliability in accordance with MIL-STD-781 and MIL-HDBK-781 using test plan V-D and the parameters θ_0 = 6000 and θ_1 = 2000. The accept-reject criteria shall be in accordance with test plan V-D of MIL-HDBK-781. The performance parameters to be monitored during the reliability testing shall be as specified in the individual tests of Table IV of this document. These parameters shall be included in the detailed reliability test procedure prepared by the supplier."

* 4.5.8.2 through 4.5.8.6: Delete text.

* Add: "4.5.9 Environmental Stress Screening (ESS). Each amplifier shall be non-energized and subjected to 25 temperature cycles from -54°C to +71°C chamber temperature at a temperature rate change of 15°C per minute minimum. Temperature dwell at temperature extremes shall be that time required for the item to stabilize in temperature. Stabilization occurs when the temperature of the part of the amplifier having the longest thermal lag has met or exceeded the maximum or minimum temperature for 5 minutes."

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